

Section B

6 **(short)** A half adder adds two single binary digits (A and B) and has two outputs, sum (S) and carry (C_{out}). A full adder circuit adds two binary digits but also accounts for a digit carried in (C_{in}) as well as out (C_{out}).

(a) Derive logical expressions for the outputs of the half adder in terms of its inputs and draw a circuit implementation using one XOR gate and other gates if needed. [4]

(b) Derive logical expressions for the outputs of the full adder in terms of its inputs and draw a circuit implementation using logic gates. [6]